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Blockchain in health supply chain management: State of art challenges and opportunities

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Abstract

Rising global costs and the fragmented nature of the healthcare supply chain creates several challenges. The healthcare industry needs efficient solutions that can streamline supply chain operations and processes in a cost- effective manner. Leading healthcare players are exploring blockchain technologies to achieve efficiencies and gain better control over their supply chains.
In this paper, we present the current state of the subject and summarize the benefits and the challenges of the distributed organization and management of supply chains management. Focusing on healthcare, and we discuss the applicability of blockchains in the supply chain domain in the healthcare.

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1. Introduction

Health organizations encounter many challenges accompanied with new requirements, namely; customer dissatisfaction, increasing cost of the health services, competition and reducing the reimbursement for services. All of these factors force the health organizations to adopt a system that can meet these requirements, dealing with the continuous changes, technology changes, increase in the health services costing, increase in competitive position and gaining customers' satisfaction [1].

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There is an increasing need by private hospitals administrators to adopt supply chain management; to promote the advancement of health care [2].

Due to cost control Supply chain management had become an important phenomenon by health care organizations in order to achieve the organization's set goals.Supply chain management (SCM) deals with different categories of flows; namely, flows of goods, flows of information and flows of funds within and among supply chain partners in order to satisfy consumer needs in the most efficient way [3]. Though effective SCM is the biggest challenge in every industry, in healthcare, there is additional risk and complexity as a compromised supply chain in healthcare can affect the patient's safety.
Blockchain in healthcare improves overall security of patients' moreover resolves the issues of drugs authenticity and drugs supply chain traceability and enables secure interoperability between healthcare organizations.
Blockchain technology applications in healthcare are segmented into supply chain management, patient data management, clinical trials & data security, drug traceability, claims adjudication, billing, and others. Among all these, the supply chain management segment is considered as one of the most promising and the most Blockchain technology applications in healthcare are segmented into supply chain management, patient data management, clinical trials & data security, drug traceability, claims adjudication, billing, and others.
The aim of this paper is to provide an overview of fundamental concepts related to the blockchain technology of healthcare supply chain management.
The remainder of this paper is organized as follows: Section two presents a background of healthcare supply chain management with a description of its key elements and an overview of the problems in the health domain. Section three we present blockchain technology. In section four we present the application of blockchain for supply chain management. Finally, section five presents a conclusion of this study, including strengths and limitations of this study.

2. Healthcare Supply chain management

A supply chain is often described as a system of organizations, people, activities, information, and resources involved in moving a good or service from the initial supplier to the final customer [4] Similar to supply chain management in a manufacturing setting, health supply chains can be characterized by different modes of integration:

- Integration and co-ordination of processes.
- Integration and co-ordination of information flows.
- Integration and co-ordination of planning processes.
- Integration of intra- and inter-organizational processes.
- Integration of market-approach.
- Integration of market-development.

However, supply chain management is a challenging prospect in healthcare; with scattered ordering settings of medical supplies, drugs and critical resources, there is inherit risk of compromising the supply chain process that might directly impact patients' safety [5, 6]
According to a study by the World Health Organization (WHO) [7], more than 100,000 people die in Africa due to improper dosing from counterfeited drugs ordered from not known or trusted vendors. In addition to product and drug counterfeiting, lacking product registry and packaging errors in a healthcare facility may disrupt the entire SCM [8].

3. Blockchain

3.1. Blockchain technology

The detailed technical underpinning of the blockchain technology is outside the scope of this paper. However, for the purpose of our discussion going forward, it is important to shed light on some blockchain concepts, features and terminologies that will foster the understanding of how blockchain is applied to solve healthcare in supply chain management problems.

By definition, blockchain is a decentralized ledger that is shared by all network participants. Because of its nature, modifying an existing ledger is not mathematically possible. This is achieved through the use of cryptographic algorithms. Blockchain data structure is a list of data blocks that are timestamped, immutable, and in strict order [9] Immutability is implemented using a hash, a digital fingerprint of data. Every block has a reference to a previous block's hash and thus gives a strict order to the blockchain. Following hashes from the current block ends with block 0—called the genesis block. It is the first created block on a

specific blockchain. Blocks contain a list of transactions. This type of data structure enables provenance, that is, a single place of origin for any transaction. Since all blockchain transactions are timestamped and immutable, fraudulent drug traffickers can be easily identified. There are two types of blockchains: Public and private depending on who has the ability to be a user of, or run a node on, the blockchain. Public blockchains (e.g., Bitcoin) allow every node in the network to conduct transactions and participate in the consensus process. On the contrary, private blockchains (e.g., Multichain) only allow a limited number of approved nodes to participate in the consensus process. The reliable healthcare companies need to register their products in private blockchain systems to ensure the reliability and quality of their products [10].

4. Blockchain for supply chain management

In healthcare supply chain management, the blockchain technology transactions is particularly key monitoring technology for tapping into the whole process of drugs and medical products movement [11].Since all transactions are recorded onto the ledger, and every node in the blockchain maintains a record of the transaction, it is easy to verify the origin of the drug, the vendor and the distributor instantly. Furthermore, the distributed ledger of the blockchain allows healthcare officials and physician to check and authenticate the credentials of suppliers [12].

With better insight into the supply chain through proper and timely authentication process, pharmacies and healthcare providers will be able to ensure that the flow of authentic drugs continues to reach those patients who need it the most. In this regard, blockchain technology holds a great promise for establishing a trusted network of vendors that allow healthcare administrators to guard patients from disreputable suppliers. Furthermore, blockchain technology promises significant enhancement on demand forecasting, data provenance, fraud prevention, and transaction. Figure 1 illustrates a pharmaceutical SCM process using blockchain technology.

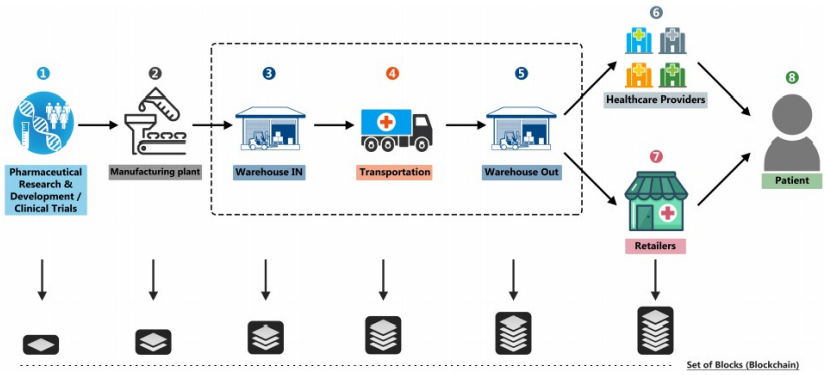

Fig. 1. Supply chain management in blockchain [13]

- Step-1: A block is created upon the invention of a new medicine or medical care which includes patent protection and a long process of clinical trials. This information is recorded in the digital ledger as a form of transaction.
- Step-2: Once the clinical trial is successful, the patent is sent to the manufacturing plant for test prototype and mass production. Every product has its own unique identity that is integrated with another transaction or block in the blockchain including other relevant information.
- Step-3: Once the mass production along with packaging is finished, medicine is gathered in a warehouse for future distribution. Information such as, time, lot number, barcode, expiry date is included in the blockchain.
- Step-4: Transportation information is also included in the blockchain which may include time out from one warehouse (IN) to other, mode of transportation, authorized agent, and other information.
- Step-5: A third-party distribution network is normally responsible for distributing drugs and medical supplies to healthcare providers or retailers. A warehouse (OUT) for each third party is used for this purpose from where all distribution endpoints are linked. A separate transaction is also integrated into the blockchain.
- Step-6: Care providers such as hospitals, or clinics need to provide information, for example, batch number, lot number, product owner, expired date to authenticate, and prevent counterfeit.This is also included in the blockchain.
- Step-7: The actions taken by a retailer are similar to Step-6.
- Step-8: Patients are encouraged to determine authenticity throughout the whole process as blockchain supply chain offers transparent information for verification to potential buyers.

5. Conclusion

A Blockchain network acts as a peer-to-peer network, where the immutable information is shared among all participants in full transparency. The one great benefit of Blockchain for healthcare supply chain is that it allows all stakeholders to verify the authenticity of the transactions stored inside the ledger.
In this paper, we investigated applications of Blockchain and their benefit to healthcare supply machine management. However the reality is also that there's a cost associated to implementing and running this technology, which for most part remains unregulated to this day. After all, the latter may actually be the biggest hurdle at the moment.

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